Neural networks trained on compact group composition learn the global form of the group

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Abstract

Small networks trained on finite group composition are known to learn matrix coefficients of a few irreducible representations. We show that the same structure emerges, in sharper form, when the group is a continuum. Across ten compact group forms drawn from $\mathrm{SU}(2)$, $\mathrm{SU}(3)$, $\mathrm{Spin}(4)$, and $\mathrm{Spin}(5)$ and their central quotients, presented through fixed generic encodings, every seed concentrates its encoder variance on the matrix coefficients of a smallest faithful representation of the task group, so half-integer spins appear exactly when the encoding distinguishes the two sheets of the double cover and the learned features determine the isomorphism type of the group. Ablations, activation patching, and a new test based on the left and right translation actions verify the mechanism and show that the features span the full matrix-coefficient space of the selected representation. Matched training on $\mathrm{SO}(3)$ explains the selection, since the 9-dimensional coefficient space beats a 6-dimensional faithful frame on margin at 17% lower parameter norm. Under regression onto heat-kernel targets, the same architecture fills isotypic subspaces in order of Casimir eigenvalue up to a truncation set by the weight decay. One finding is negative. On $\mathrm{SO}(5)$, a stable vector-plus-adjoint mixture survives every replication we ran, so the minimal-representation description holds in nine of the ten group forms and fails in one.

1 Introduction

When a small network learns to compose rotations, which group does it learn? The rotation group $\mathrm{SO}(3)$ and its double cover $\mathrm{SU}(2)$ share a Lie algebra, yet they are different groups with different irreducible representations. For finite groups, the corresponding question is answered in detail. Networks trained on a group multiplication table learn matrix coefficients of a few irreducible representations and score candidates through characters [Chughtai et al., 2023, Sec. 4], and all of this work presupposes a finite table and one learned embedding per element. Whether the representation theory emerges in a generic network when the group is a continuum, where no embedding table can exist, has remained open.

In this work, we answer that question for compact Lie groups. We train encoder-factorized scorers to identify products in ten group forms drawn from $\mathrm{SU}(2)$, $\mathrm{SU}(3)$, $\mathrm{Spin}(4)$, and $\mathrm{Spin}(5)$ and their central quotients, with each element presented through a fixed generic encoding, and we verify our understanding with four lines of evidence, namely concentration of the encoder on the matrix coefficients of a single faithful representation, causal ablations in both directions, translation operators and activation patching, and character read-off of the score.

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The learned representation determines the global form of the group. Averaging the encoding over a central subgroup Z realizes the same task on the quotient G/Z, so the isomorphism type becomes an experimental variable. In every seed of every condition, the model concentrates 93–98% of encoder variance on a smallest faithful representation of the task group, and half-integer spins appear exactly when the encoding distinguishes $\pm g$ (Section 6). Unlike the finite-group setting Chughtai et al. [2023], selection is deterministic across seeds.

The learned features span the full matrix-coefficient space, and cost explains why. The active subspace has exactly the real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ that the real Schur lemma prescribes, and a new two-sided translation test certifies the full space, which the trained features pass while a 6-dimensional faithful frame fails linear right closure (Section 5). In matched training, the 9-dimensional coefficient space beats the frame on margin at 17% lower parameter norm, so regularized training selects the cheapest complete mechanism.

Regression reveals a spectral bias in order of the Casimir eigenvalue. Ranking needs one character, so it hides the network's allocation policy. Regression onto heat-kernel targets requires many isotypic subspaces at known weights, and the model fills them in order of Casimir eigenvalue up to a sharp truncation that moves with the weight decay (Section 7).

One finding is negative. On $\mathrm{SO}(5)$, a stable vector-plus-adjoint mixture survives five seeds and every conditioning variation we ran, so the minimal-representation description fails as stated in one of ten group forms (Section 6).

Paper organization. Sections 2 and 3 cover related work and the setup, Section 4 the character mechanism, Section 5 the mainline reverse engineering, Sections 6 and 7 the ten-group and spectral-bias studies, and Section 8 concludes. The appendix carries the full record, and Appendix A maps it.[1]

2 Related work

Chughtai et al. [2023] reverse engineer networks trained on finite group composition and propose the group composition via representations (GCR) algorithm, in which embeddings hold representation matrices and logits are characters, Stander et al. [2024] argue for a coset-based description of the same models, and Wu et al. [2025] reconcile the two accounts. All of this work presupposes a finite table and free embeddings, the setting of the grokking literature Power et al. [2022], Nanda et al. [2023], Gromov [2023], Zhong et al. [2023], the associated theory proves emergence of irreducible features for finite groups Morwani et al. [2024], Marchetti et al. [2024], and equivariant architectures build the harmonics in by construction Cohen and Welling [2016], Geiger and Smidt [2022]. Our object differs in kind. A continuum group admits no embedding table, every element can be held out, and the global form of the group becomes an experimental variable. Rahaman et al. [2019] name the spectral bias toward low frequencies, which we quantify on a group manifold. Appendix B details the comparisons and the safeguards we adopt from the adjudication literature.

3 Setup and background

Task. Given encodings of $a, b \in G$ and of $K = 64$ candidates, the model must identify the product. The positive candidate is $(ab)\exp(\xi)$ with ξ a centered Gaussian of scale 0.2 on the Lie algebra, the 63 negatives are Haar samples excluded from a ball around the positive class, and the loss is cross-entropy over the K scores. Training pairs come from a fixed pool of 2048 Haar elements at training fraction $f = 0.01$ unless stated, and we report ranking accuracy on pairs of fresh elements never seen in training (Appendix C).

Encodings and quotient conditions. Each element enters through a frozen map Ψ, so the model has no per-element parameters. The mainline encoding evaluates 32 fixed Haar landmark kernels, standardized under Haar statistics. For a central subgroup $Z \subseteq Z(G)$, the quotient condition replaces Ψ by its average over the Z-orbit, which is exactly Z-invariant and separates the elements of G/Z, so it realizes the same task on the quotient group with all hyperparameters shared. Every headline result reproduces under two further encodings (Appendix G).

[1] Code, tracked results, and the scripts that regenerate every table are included as anonymized supplementary material.

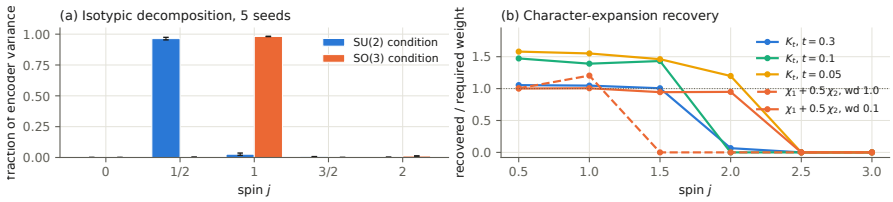


Figure 1: (a) Fraction of encoder variance per isotypic subspace for the mainline conditions, five seeds. The conditions differ only in whether the encoding distinguishes $\pm g$, and each concentrates on the smallest faithful irrep of its own group. (b) Recovery of the character expansion of the learned score for heat-kernel and band-limited targets. Values near 1 mean a subspace carried at its required weight, and the dashed curve shows the in-support spin-2 subspace pruned at weight decay 1.0 and recovered at 0.1 (Section 7).

Model. An encoder-factorized scorer with about 0.3M parameters. A shared two-layer GELU encoder produces activations $z(a), z(b) \in \mathbb{R}^{128}$, an interaction network produces $h(a, b)$, a candidate encoder produces $z_c(c)$, and the score is $s(a, b, c) = \langle h(a, b), z_c(c) \rangle$, trained by AdamW with weight decay 1.0. We reverse engineer one mainline model per condition and replicate every quantity over five seeds (mean $\pm$ standard deviation).

Mathematical background. A representation of a compact group G is a continuous homomorphism $\rho\colon G \to GL(V)$, irreducible if V has no proper invariant subspace, and faithful if ρ is injective (we use the shorthand irrep). The character is $\chi_\rho(g) = \operatorname{tr} \rho(g)$, and the functions $g \mapsto \rho(g)_{ij}$ are the matrix coefficients of ρ. By the Peter–Weyl theorem, the matrix coefficients of all irreps span a dense subspace of $L^2(G)$ and are orthogonal across irreps, so any square-integrable feature of the group decomposes into isotypic subspaces indexed by irreps. Our networks compute with real numbers, the commutant $\operatorname{End}_G(W)$ of an irreducible real representation W is $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$, and the real coefficient space of W has dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \operatorname{End}_G(W)$. The center acts in each irrep by a central character, the irreps of G/Z are exactly those with trivial central character on Z, and a Z-invariant encoding therefore forbids every subspace charged under Z (the descent rule). Appendix D develops the background, including the concrete representation theory of the five groups.

4 The character mechanism on compact groups

The GCR algorithm of Chughtai et al. [2023] extends verbatim to a compact group. Hold matrix coefficients $\rho(a)$ and $\rho(b)$ of a faithful irrep in the encoder, compute the product $\rho(a)\rho(b) = \rho(ab)$ in the interaction network, and score candidate c by the character, using the identity

$$\chi_\rho(ab\,c^{-1}) = \operatorname{tr}\big(\rho(a)\rho(b)\rho(c)^*\big) = \big\langle \rho(c),\, \rho(a)\rho(b) \big\rangle_{\mathrm{HS}}.$$

Composition is bilinear and scoring is linear in the candidate's coefficients, so the mechanism fits a factorized scorer (proofs of the two results below are in Appendix E).

Lemma 1. *Let V be a faithful finite-dimensional unitary representation of a compact group H with character χ_V. Then $\operatorname{Re}\chi_V(g) \le \chi_V(e)$ for all $g \in H$, with equality only at $g = e$, and near the identity $\chi_V(e) - \operatorname{Re}\chi_V(\exp X)$ is a positive definite quadratic form in X up to higher order.*

Lemma 1 says that a monotone function of $\operatorname{Re}\chi_V(ab\,c^{-1})$ ranks the true product above every excluded negative for any faithful V, and a representation with nontrivial kernel assigns the product and its kernel translates identical scores, so it cannot separate them (Lemma 3, Appendix E). A trained model therefore needs a faithful representation of exactly the task group and is free to select the cheapest one. Whether it stores a full coefficient space or a smaller separating frame is an empirical question, which Lemma 2 turns into a falsifiable test.

5 Reverse engineering the mainline models

Every claim below replicates in five seeds per condition, and Appendix G documents each experiment in full.

The encoder learns the smallest faithful irrep, and the double cover is visible. The $\mathrm{SU}(2)$ encoder places 0.84 ± 0.01 of its variance in the spin-$\frac{1}{2}$ isotypic subspace, with the quaternion linearly decodable at held-out $R^2 = 0.995$. The $\mathrm{SO}(3)$ encoder places 0.92 ± 0.01 in spin 1, every half-integer subspace sits at the 0.2% sampling floor, and the quaternion is not decodable. The $\mathrm{SU}(2)$ features are odd under the center, with $\cos(z(g), z(-g)) = -0.72 \pm 0.02$, and the $\mathrm{SO}(3)$ features are exactly even (Figure 1a), so the learned features see the two sheets of the double cover.

Ablations give a causal double dissociation. We ablate by subtracting from z the part linearly predictable from a chosen subspace's exact matrix coefficients. Ablating spin $\frac{1}{2}$ collapses the $\mathrm{SU}(2)$ model to chance (0.026 ± 0.002 against chance 0.016) and collapses its center discrimination to the tie value, and is a no-op on the $\mathrm{SO}(3)$ model (1.000 ± 0.000). Ablating spin 1 transposes the pattern exactly. Random subspaces of equal rank are inert in both conditions, and restricting to the claimed subspace preserves performance (Table 4). The learned decision rule is the compact-group form of the GCR logit, with the fundamental character alone explaining 0.97 ± 0.01 of score variance for $\mathrm{SU}(2)$ and 0.88 ± 0.01 for $\mathrm{SO}(3)$ (Experiment 5).

Translation operators realize the representation, and patching confirms sufficiency. Fitted affine maps T_k with $z(kg) \approx T_k z(g)$ reach held-out $R^2 = 0.90$–0.98, and on the dominant subspace they match d_j copies of the irrep, with eigenvalue phases agreeing to 0.0014 ± 0.0001 radians for $\mathrm{SU}(2)$ (Experiment 6). Replacing the internal product state $h(a, b)$ by its reconstruction from the eight realified spinor coefficients of a counterfactual product $a'b'$ redirects the model to select $a'b'$ with probability 0.996–0.999, while the wrong-irrep reconstruction redirects at chance, and the interaction output concentrates on the span of the matrix product $W(a)W(b)$, so the network computes the representation's own composition law (Experiment 7).

The learned features span the full coefficient space, which the two-sided test certifies. In every seed, the active subspace of z has exactly the real dimension of the full coefficient space, 4 for the quaternionic spinor and 9 for the real-type vector irrep. The following lemma turns full-space identity into a falsifiable prediction.

Lemma 2. *For an irreducible unitary representation ρ of a compact group G, the span $M_\rho \subseteq L^2(G)$ of its matrix coefficients is irreducible under $G \times G$ acting by left and right translation, so every nonzero subspace of M_ρ stable under both actions equals M_ρ. For an irreducible real representation W with commutant $D = \mathrm{End}_G(W)$, the real coefficient space has dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} D$ and is the minimal nonzero two-sided translation-stable real subspace of its isotypic component.*

On the full coefficient space, left and right translation both act linearly and the actions commute, and on any proper subspace at least one action fails. Fitted operators on the trained $\mathrm{SO}(3)$ features reach held-out $R^2 = 0.989 \pm 0.001$ per side with commutator defect 0.012 ± 0.001, matching the exact coefficient space, while the 6-dimensional two-column frame passes the left test exactly and fails right closure at $R^2 = 0.638$, and a random 6-dimensional truncation fails both (Table 7).

Matched training shows the full space is the cheaper mechanism. The frame determines the rotation with 6 coordinates, so storage minimality cannot explain the selection. In matched training on exact frame features against exact coefficient features, the coefficient arm attains margin 7.29 ± 0.16 against 6.49 ± 0.05 at statistically equal accuracy and 17% lower parameter norm (Table 8). The frame arm reconstructs its missing column nonlinearly, and that costs more than the three saved coordinates, so regularized training selects the full coefficient space as the cheapest complete mechanism.

6 Universality across ten group forms

Selection is deterministic and follows the descent rule. Ten group forms test whether the mainline findings are general (Table 1). In nine of ten, the encoder concentrates on a smallest faithful representation of the task group, the same one in all five seeds, with every subspace the descent rule forbids at the floor. On $\mathrm{SU}(3)$, the measured center cosine is -0.484 ± 0.005 against the exact prediction $\cos(2\pi/3) = -0.5$. When the center is not cyclic, no faithful irrep exists (Fact 1), and the $\mathrm{Spin}(4)$ models build the direct sum that faithfulness forces. Where a smaller reducible sum competes against a larger faithful irrep, the sum wins in every seed.

Each summand is necessary for exactly the discriminations its central character supports. The mainline double dissociation repeats under the order-three center of $\mathrm{SU}(3)$ (Table 5), and on $\mathrm{Spin}(4)$,

Table 1: The learned representation across ten group forms, with the fraction of encoder variance on its matrix coefficients (mean over five seeds) and the rejected faithful alternative where one exists. $\mathrm{Spin}(4)$ irreps are labeled (j_L, j_R). Full spectra, seed statistics, and interventions are in Appendices H and J.

task group	learned representation	variance	rejected alternative
$\mathrm{SU}(2)$	spin $\frac{1}{2}$	0.96	
$\mathrm{SO}(3)$	spin 1	0.98	
$\mathrm{SU}(3)$	$\mathbf{3} \oplus \bar{\mathbf{3}}$	0.97	
$\mathrm{PSU}(3)$	adjoint $\mathbf{8}$	0.93	
$\mathrm{Spin}(4)$	$(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	0.98	(no faithful irrep exists)
$\mathrm{SO}(4)$	$(\frac{1}{2}, \frac{1}{2})$	0.97	
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	$(1, 0) \oplus (0, \frac{1}{2})$, 5 dims	0.98	$(1, \frac{1}{2})$, 6 dims, at 0.008
$\mathrm{PSO}(4)$	$(1, 0) \oplus (0, 1)$, 6 dims	0.98	$(1, 1)$, 9 dims, at 0.011
$\mathrm{Spin}(5)$	spinor $\mathbf{4}$	0.97	
$\mathrm{SO}(5)$	vector $\mathbf{5}$ + adjoint $\mathbf{10}$	$0.69 + 0.22$	(negative result)

whose center supports three distinct discriminations, every cell of the summand-resolved intervention map lands where the central character puts it (Table 6).

The exception is $\mathrm{SO}(5)$, and we report it as a failure. The $\mathrm{SO}(5)$ encoder holds the vector $\mathbf{5}$ at 0.68–0.70 and the adjoint $\mathbf{10}$ at 0.21–0.23 of variance, stable across five seeds, doubled training, four times the candidates, and reduced weight decay, and no condition solves the task. Two hypotheses remain open, purification as the task approaches a solution, or a genuinely mixed minimum-cost mechanism, which the cost result of Section 5 makes coherent (Experiment 12).

7 Spectral bias in order of the Casimir eigenvalue

We regress the same architecture onto the $\mathrm{SU}(2)$ heat kernel $K_t(g) = \sum_j (2j+1) e^{-tj(j+1)} \chi_j(g)$ evaluated at $ab\,c^{-1}$ and measure the recovered character expansion of the learned score (Figure 1b, Appendix I).

The learned score is a truncated character expansion that fills in Casimir order. Recovered subspaces sit at ratio near 1 and subspaces beyond a sharp truncation sit at 0, with the truncation at higher Casimir eigenvalue for smaller t and advancing in Casimir order during training, so that at $t = 0.05$ the spin-2 recovery climbs from 0.12 to 1.20 while spin $\frac{5}{2}$ stays at 0.

The truncation is a cost boundary set by regularization. On the band-limited target $\chi_1 + \frac{1}{2}\chi_2$, weight decay 1.0 prunes the spin-2 part, and the abandoned 12.6% of target variance equals the final test error. Weight decay 0.1 recovers the target with profile $(1.00, 1.01, 0.94, 0.95)$ and out-of-support subspaces at 0.000, which also calibrates the measurement. A subspace survives when its variance contribution exceeds its norm cost, the cost-boundary logic of Section 5, and the center-invariant target forces integer spins even on the faithful $\mathrm{SU}(2)$ encoding (Experiment 15).

8 Discussion and conclusion

Small generic networks trained on compact group composition learn the irreducible representations of the exact global form the data determine, store them as full matrix-coefficient spaces because that is the cheapest complete mechanism, and allocate spectral capacity in Casimir order under a regularization-set truncation. The results suggest a selection principle we can state but not yet prove, that among near-optimal parameter settings the minimum-norm one realizes the full coefficient space of a cost-minimizing faithful representation with a monotone character score (Appendix K states the conjecture precisely).

Limitations and future work. Every quotient condition generalizes below its cover, and we have not yet separated representation selection from the conditioning of the quotient encoding. Causal interventions carry weaker positive than negative evidence Wu et al. [2025], which we mitigate with the two-sided test, exact no-op controls, and matched retraining. Solving $\mathrm{SO}(5)$ would separate the two open hypotheses, and proving the conjecture would give the first selection theorem for a continuous group.

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A Structure of the appendix

Appendix B details the relation to prior work on group composition. Appendix C specifies the task, encodings, architecture, and training. Appendix D develops the representation theory and harmonic analysis the paper uses, including the concrete representation theory of the five groups. Appendix E proves the three results. Appendix F documents the analysis methods in the detail needed to reimplement them, including the failure mode that motivated the function-space ablation. Appendix G is the

experimental record for the mainline $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ program, organized as Experiments 1–9. Appendix H covers the remaining group forms as Experiments 10–12, and Appendix I the heat-kernel program as Experiments 13–15. Appendix J collects the summary tables referenced from the main text, and Appendix K answers questions the record raises.

B Relation to prior work on group composition

What transfers from the finite-group analyses. The GCR description of Chughtai et al. [2023] has three components, embeddings that hold representation matrices, an interaction that multiplies them, and a character read-off. All three transfer to our setting and all three are verified here, with matrix coefficients of the continuous group in place of finite representation matrices (Experiments 1, 4–7). The character identity behind the read-off is the same Hilbert–Schmidt pairing in both settings.

What has no finite analogue. A finite model owns one embedding row per group element, so its representation content is a property of a learned table, and the analysis cannot hold elements out of the representation itself. Our elements enter through a frozen map, the encoder is a function on the group manifold, and all of its structure generalizes to fresh Haar samples by construction. Two consequences matter. First, T2 evaluation on never-seen elements (Appendix C) has no finite-group counterpart. Second, the quotient construction, which averages the encoding over a central subgroup, turns the global form of the group into an experimental variable with the Lie algebra held fixed. Nothing analogous exists for a finite multiplication table presented through free embeddings, because free embeddings carry no topology to quotient.

Determinism against arbitrariness. Chughtai et al. [2023] find that which irreps a finite-group model learns varies by seed, and Zhong et al. [2023] find qualitatively different algorithms under small hyperparameter changes. Our selection is deterministic in all fifty mainline-program seeds and all forty universality seeds. A plausible reason is that $\mathrm{SU}(2)$ has one irrep per dimension, so the equivalent-choice degeneracy that drives finite-group arbitrariness is absent. We state this as an observation, and the determinism itself is one of our findings.

The adjudication lesson. Stander et al. [2024] and Wu et al. [2025] show that observational evidence on these tasks underdetermines the mechanism, and Wu et al. [2025] show that causal interventions can pass on models where the tested explanation cannot hold. We adopt three safeguards. The quotient conditions provide exact no-op controls, interventions that provably act as the identity and measure as such (Experiments 4 and 10). The two-sided translation test is a structural discriminator with competing feature systems evaluated under the same fit (Experiment 8). The frame-versus-coefficient comparison replaces inference from a trained model by matched retraining (Experiment 9).

C Task, encodings, architecture, and training

Task construction. A dataset is built from a pool of $N = 2048$ Haar elements of G. A training example is an ordered pair (a, b) from a frozen random fraction f of the N^2 pairs, together with $K = 64$ candidates, namely one positive $(ab)\exp(\xi)$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$ on the Lie algebra at $\sigma = 0.2$, and 63 negatives drawn from Haar measure and resampled until their distance to the positive class exceeds 0.5. Candidates are frozen at dataset construction. The loss is cross-entropy over the K scores. Under this construction the Bayes-optimal rule ranks by distance to ab, with Monte-Carlo accuracy 0.951 for $\mathrm{SU}(2)$ at $K = 64$, falling to 0.936 at $K = 128$ and 0.922 at $K = 256$.

Evaluation tiers. T1 is ranking accuracy on held-out pairs of pool elements, the analogue of held-out cells of a multiplication table. T2 is accuracy on pairs of fresh Haar elements never seen in training, which has no finite-group analogue and is the headline metric. T3 places the 63 negatives at a fixed geodesic radius r from the product and profiles resolution as a function of r.

Encodings. The mainline encoding is $\Psi(g)_i = \exp(\alpha\langle g, h_i\rangle)$, $i = 1, \ldots, 32$, with fixed Haar landmarks h_i and $\alpha = 3$, standardized to zero mean and unit variance under Haar statistics. The two robustness encodings are a frozen random two-layer tanh network of the raw coordinates, and the values of a fixed random band-limited function at 32 translated probe points. Quotient conditions replace the encoding by its average over the quotiented central subgroup, namely the antipodal aver-

age for $\mathrm{SO}(3)$ and $\mathrm{SO}(5)$, the order-three orbit average for $\mathrm{PSU}(3)$, and the three sign-pair averages for the quotients of $\mathrm{Spin}(4)$. Each average is exactly invariant under the quotiented subgroup and distinguishes all other center elements, which we verify numerically at construction.

Architecture. The shared argument encoder is a two-layer GELU network $\mathbb{R}^{32} \to \mathbb{R}^{256} \to \mathbb{R}^{128}$ producing $z(a)$ and $z(b)$. The interaction network is an MLP on the concatenation, $\mathbb{R}^{256} \to \mathbb{R}^{512} \to \mathbb{R}^{128}$, producing $h(a, b)$. A separate candidate encoder with the encoder architecture produces $z_c(c)$, and the score is the inner product $\langle h(a, b), z_c(c) \rangle$. The model has about 0.3M parameters.

Training. AdamW with learning rate 10^{-3}, weight decay 1.0, batch size 4096 (2048 for $\mathrm{SU}(3)$, $\mathrm{Spin}(4)$, and $\mathrm{Spin}(5)$), and 100k–200k steps, with $f = 0.01$ unless stated. Heat-kernel and band-limited regressions replace the cross-entropy by mean squared error on standardized targets, with candidates drawn half near the product at uniform geodesic radius and half from Haar measure. All runs use a single NVIDIA T4 GPU, PyTorch 2.11, and complete in hours (mainline conditions) to about two days (the largest groups). Five seeds per reported condition throughout.

Groups and exact coefficients. Elements of $\mathrm{SU}(2)$ are unit quaternions, of $\mathrm{SU}(3)$ Haar-random special unitary matrices, of $\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ pairs of unit quaternions, and of $\mathrm{Spin}(5) = \mathrm{Sp}(2)$ quaternionic 2×2 unitaries sampled by quaternionic Gram–Schmidt. Matrix coefficients for half-integer spins of $\mathrm{SU}(2)$ are computed exactly through the action on homogeneous polynomials, for $\mathrm{SU}(3)$ through symmetric powers of the fundamental and the adjoint action on the Gell-Mann frame, and for $\mathrm{Sp}(2)$ through the complex embedding together with the conjugation actions on Hermitian and anti-Hermitian quaternion matrices. The primitives pass a unit-test suite covering the homomorphism property, unitarity, characters, central characters, Schur orthogonality, planted-component recovery, and exact invariance of the quotient encodings.

Reproducibility. A tracked results bundle records, for every run, the configuration, seed, checkpoint hash, final metrics, and analysis outputs, and every table in this paper regenerates from the bundle. Code and the bundle are included as anonymized supplementary material.

D Background: representation theory and harmonic analysis on compact groups

This appendix collects the facts the paper uses, with references in place of proofs. Standard sources are Bröcker and tom Dieck [1985] for compact groups and the Peter–Weyl theorem, Sepanski [2007] for the real Schur lemma and the type classification, Folland [1995] for harmonic analysis, and Fulton and Harris [1991] for the representation theory of the classical groups. Throughout, G is a compact Lie group and all representations are finite-dimensional and continuous.

Haar measure and unitarity. A compact group carries a unique translation-invariant Borel probability measure, its Haar measure, and every expectation over group elements in this paper is taken with respect to it. Averaging an arbitrary inner product over the group makes any representation unitary, and every unitary representation decomposes as a finite orthogonal direct sum of irreducible ones [Bröcker and tom Dieck, 1985, Ch. II]. The set of isomorphism classes of complex irreps is denoted $\widehat{G}$, and for $\rho \in \widehat{G}$ we write $d_\rho = \dim V_\rho$ and $\chi_\rho(g) = \operatorname{tr} \rho(g)$ for its character.

Matrix coefficients and orthogonality. For $\rho \in \widehat{G}$ and vectors $u, v \in V_\rho$, the function $g \mapsto \langle u, \rho(g)v \rangle$ is a matrix coefficient of ρ. Schur's lemma states that a linear map commuting with two irreducible representations is zero or an isomorphism, and that the self-intertwiners of a complex irrep are the scalars [Fulton and Harris, 1991, Ch. I]. Its integrated form is the orthogonality of coefficients. In a unitary basis,

$$\int_G \rho(g)_{ij}\, \overline{\sigma(g)_{kl}}\, dg \;=\; \frac{\delta_{\rho\sigma}\, \delta_{ik}\, \delta_{jl}}{d_\rho}, \qquad \int_G \chi_\rho(g)\, \overline{\chi_\sigma(g)}\, dg \;=\; \delta_{\rho\sigma}$$

for $\rho, \sigma \in \widehat{G}$ [Bröcker and tom Dieck, 1985, Ch. II]. These relations make the coefficient spans of distinct irreps orthogonal in $L^2(G)$, so the per-subspace variance fractions of Appendix F are well defined, and our implementation verifies them numerically on the sampled probe sets.

The Peter–Weyl theorem. The matrix coefficients of all $\rho \in \widehat{G}$ span a dense subspace of $L^2(G)$, and as a representation of $G \times G$ acting by left and right translation,

$$L^2(G) \;\cong\; \bigoplus_{\rho \in \widehat{G}} M_\rho, \qquad M_\rho \;\cong\; V_\rho^* \boxtimes V_\rho,$$

an orthogonal Hilbert-space direct sum in which each coefficient space M_ρ has dimension d_ρ^2 [Bröcker and tom Dieck, 1985, Ch. III]. Every function a network computes on G therefore has a spectral decomposition over $\widehat{G}$, and the structural question of this paper, which subspaces a trained model occupies and whether it occupies them fully, is a question about this decomposition. Lemma 2 records the two-sided irreducibility of M_ρ and its real form.

Real representations and Schur type. The networks compute with real numbers, so the relevant decomposition is that of real representations. For an irreducible real representation W, the real Schur lemma states that the commutant $\mathrm{End}_G(W)$ is a real division algebra, hence isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$ [Sepanski, 2007, Ch. III], and the three cases correspond to the classical trichotomy for a complex irrep V. The irrep V has real type when it admits an invariant real structure, quaternionic type when it admits an invariant quaternionic structure, and complex type when $V \not\cong \bar{V}$. Realification acts on dimensions as follows. A real-type V of dimension d yields $\dim_{\mathbb{R}} W = d$ with commutant $\mathbb{R}$. A complex-type or quaternionic-type V of dimension d yields $\dim_{\mathbb{R}} W = 2d$ with commutant $\mathbb{C}$ or $\mathbb{H}$, and in the complex case V and $\bar{V}$ realify to the same W, which is why a complex irrep and its conjugate span a single real isotypic subspace in our measurements. In all three cases the real coefficient space has dimension

$$\dim_{\mathbb{R}} M_W \;=\; \frac{(\dim_{\mathbb{R}} W)^2}{\dim_{\mathbb{R}} \mathrm{End}_G(W)},$$

which evaluates to d^2, $2d^2$, and d^2 in the real, complex, and quaternionic cases. The three dimensions measured in the paper are instances, namely 9 for the real-type vector irrep of $\mathrm{SO}(3)$, 18 for the complex-type fundamental pair of $\mathrm{SU}(3)$, and 4 for the quaternionic spinor of $\mathrm{SU}(2)$.

Characters and the scoring identity. Characters are conjugation-invariant, and the irreducible characters form an orthonormal basis of the class functions in $L^2(G)$ [Bröcker and tom Dieck, 1985, Ch. II]. Two identities are used throughout, namely $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$ and, for unitary ρ,

$$\chi_\rho(ab\,c^{-1}) \;=\; \mathrm{tr}\big(\rho(a)\rho(b)\rho(c)^*\big) \;=\; \big\langle \rho(c),\, \rho(a)\rho(b) \big\rangle_{\mathrm{HS}},$$

so a character evaluated at $ab\,c^{-1}$ is a Hilbert–Schmidt pairing of the candidate's coefficients against the product of the arguments' coefficients. This identity is what makes composition bilinear and scoring linear on a coefficient space, and it is the form in which the finite-group GCR mechanism extends to the compact case.

Central characters and descent to quotients. By Schur's lemma, the center $Z(G)$ acts in any $\rho \in \widehat{G}$ by a character $\omega_\rho \colon Z(G) \to U(1)$, called the central character. For a closed central subgroup Z, the irreps of the quotient G/Z are exactly those ρ with ω_ρ trivial on Z, and the spectral support of a Z-invariant function on G consists of such ρ alone. This is the descent rule. The orbit-averaged encoding is Z-invariant, so every isotypic subspace carrying a nontrivial central character of Z is forbidden, and the measured floors on those subspaces are the empirical form of the rule. One consequence deserves a proof, since it forces the direct-sum selections of Table 1.

Fact 1. *If a compact group admits a faithful irreducible complex representation, its center is isomorphic to a closed subgroup of $U(1)$. In particular, $\mathrm{Spin}(4)$, with center $\mathbb{Z}_2 \times \mathbb{Z}_2$, has no faithful irrep.*

Proof. Under a faithful irreducible ρ, the center acts by the scalars ω_ρ, and faithfulness makes $z \mapsto \omega_\rho(z)$ injective, so $Z(G)$ embeds in $U(1)$. Every finite subgroup of $U(1)$ is cyclic, and $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not. □

The Casimir element and the heat kernel. Fix a bi-invariant metric on G. The quadratic Casimir element of the universal enveloping algebra acts in each irrep ρ as a scalar $C(\rho) \geq 0$, and the matrix coefficients of ρ are eigenfunctions of the Laplace–Beltrami operator on G with eigenvalue $C(\rho)$ [Sepanski, 2007, Ch. 5]. The Casimir eigenvalue therefore plays the role of a squared frequency, and ordering isotypic subspaces by Casimir eigenvalue is the compact-group analogue of ordering Fourier modes by frequency. The heat kernel at the identity-based variable g is the class function

$$K_t(g) \;=\; \sum_{\rho \in \widehat{G}} d_\rho \, e^{-t\,C(\rho)} \, \chi_\rho(g),$$

the fundamental solution of the heat equation on G [Folland, 1995, Ch. 1], and its envelope $e^{-t\,C(\rho)}$ determines the required subspace weights in Experiment 15.

The groups of the experiments. The irreps of $\mathrm{SU}(2)$ are the spin representations D^j of dimension $2j+1$ for $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, with Casimir eigenvalue $C(D^j) = j(j+1)$, quaternionic type for half-integer j and real type for integer j, and central character $(-1)^{2j}$ at $-1 \in \mathrm{SU}(2)$ [Bröcker and tom Dieck, 1985, Ch. II]. The quotient $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm 1\}$ therefore carries exactly the integer spins, which yields the double-cover diagnostic. The irreps of $\mathrm{SU}(3)$ are labeled by highest weights (p, q) with dimension $\frac{1}{2}(p+1)(q+1)(p+q+2)$ and central character ω^{p-q} at the central element ωI for ω a primitive cube root of unity, so descent to $\mathrm{PSU}(3)$ requires $p \equiv q \pmod 3$. The subspaces tracked in the experiments are the fundamental pair $\mathbf{3} \oplus \bar{\mathbf{3}}$ at $(1,0)$ and $(0,1)$, the adjoint $\mathbf{8}$ at $(1,1)$, and the pairs $\mathbf{6} \oplus \bar{\mathbf{6}}$ and $\mathbf{10} \oplus \bar{\mathbf{10}}$ [Fulton and Harris, 1991, Ch. 13]. The group $\mathrm{Spin}(4)$ is isomorphic to $\mathrm{SU}(2) \times \mathrm{SU}(2)$, its irreps are the outer products (j_L, j_R) with central character $\big((-1)^{2j_L}, (-1)^{2j_R}\big)$ on the center $\mathbb{Z}_2 \times \mathbb{Z}_2$, and its three central quotients are $\mathrm{SO}(4)$ from the diagonal $\mathbb{Z}_2$, retaining $j_L + j_R \in \mathbb{Z}$, the product $\mathrm{SO}(3) \times \mathrm{SU}(2)$ from one factor's $\mathbb{Z}_2$, and $\mathrm{PSO}(4)$ from the full center, retaining $j_L, j_R \in \mathbb{Z}$. The group $\mathrm{Spin}(5)$ is isomorphic to the quaternionic unitary group $\mathrm{Sp}(2)$, with quaternionic spinor irrep $\mathbf{4}$, real vector irrep $\mathbf{5}$, and adjoint $\mathbf{10}$, and $\mathrm{SO}(5) = \mathrm{Spin}(5)/\{\pm 1\}$ retains the irreps with trivial central character, among them $\mathbf{5}$ and $\mathbf{10}$ [Fulton and Harris, 1991, Ch. 16].

E Proofs

Proof of Lemma 1. Each eigenvalue of the unitary matrix $V(g)$ lies on the unit circle, so $\mathrm{Re}\,\mathrm{tr}\,V(g) \leq \dim V$, with equality exactly when every eigenvalue equals 1, that is, when $V(g)$ is the identity. By faithfulness, this occurs only at $g = e$. For the expansion, write $A = dV(X)$, an anti-Hermitian matrix with eigenvalues $i\lambda_1, \dots, i\lambda_{\dim V}$. Then $\mathrm{Re}\,\mathrm{tr}\exp(tA) = \sum_k \cos(t\lambda_k) = \dim V - \frac{t^2}{2}\sum_k \lambda_k^2 + O(t^4)$, and $\sum_k \lambda_k^2 = \|A\|_F^2$. The trace of A is purely imaginary and contributes nothing to the real part. If $\|dV(X)\|_F = 0$, then the one-parameter subgroup $\exp(tX)$ lies in the kernel of V, which is trivial, so $X = 0$, proving positive definiteness. □

Lemma 3 (Kernel collisions)**.** *Let H be a compact group. Assume the score factors as $s(a,b,c) = S(\varphi(a), \varphi(b), \varphi(c))$ where φ factors through the matrix coefficients of a representation V of H with nontrivial closed kernel N. Then $s(a,b,ab\,n) = s(a,b,ab)$ for every $n \in N$ and all (a,b), so the scorer assigns identical scores to the exact product and to each of its kernel translates, and every center discrimination supported on N sits exactly at its tie value.*

Proof. Every matrix coefficient of V is constant on N-cosets, so φ is, and $ab\,n$ lies in the N-coset of ab. □

Exact collisions do not by themselves bound the population ranking loss away from its optimum. With a finite kernel and a continuous candidate distribution, sampled negatives avoid the collision set with probability one, and a uniform gap would additionally require a modulus of continuity for the scorer class, a density lower bound near the kernel translates, and a quantified comparison with the noisy positive. We leave that quantitative statement open and use only the collision form, which is exactly what the tie-aware center discriminations measure.

Proof of Lemma 2. Fix the conventions $(L_k f)(g) = f(k^{-1}g)$ and $(R_k f)(g) = f(gk)$, so that for the coefficient $f_{u,v}(g) = \langle u, \rho(g)v\rangle$ left translation acts on the u slot through the contragredient and right translation acts on the v slot. The map $V_\rho^* \otimes V_\rho \to L^2(G)$ sending $u^* \otimes v$ to

$f_{u,v}$ then intertwines the $G \times G$-action with the outer tensor product action and is injective by the orthogonality of matrix coefficients (Appendix D). Over $\mathbb{C}$, the outer tensor product of irreducible representations is an irreducible representation of the product group, so M_ρ has no proper nonzero two-sided invariant subspace, and $\dim_{\mathbb{C}} M_\rho = (\dim V_\rho)^2$. For the real form, the real Schur lemma gives $D = \mathrm{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and by the Jacobson density theorem the algebra spanned by $\{W(g)\}_{g \in G}$ is all of $\mathrm{End}_D(W) \cong M_m(D^{\mathrm{op}})$ with $m = \dim_D W$, of real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} D$. The real coefficient span of W is the image of $\mathrm{End}_D(W)$ under the trace pairing, which identifies it with $W^* \otimes_D W$ at that dimension. A real subspace of the coefficient span invariant under both translation actions corresponds to a two-sided ideal of $\mathrm{End}_D(W)$, and $M_m(D^{\mathrm{op}})$ is a simple algebra, so the only nonzero such subspace is the whole span. □

F Analysis methods in detail

Isotypic FVE. For a fixed probe set of 2048 Haar elements, we compute the fraction of variance explained (FVE) of the centered activations $z(g)$ by the sampled spans of exact matrix coefficients, one isotypic subspace at a time. The sampled spans of different subspaces are jointly orthonormalized by QR in order of Casimir eigenvalue, which bounds cross-subspace leakage, and the residual finite-sample leakage puts the measurement floor near 0.2% of variance. We verified the floor on white noise and on planted single-subspace features, which the method recovers exactly.

Function-space ablation. To ablate an isotypic subspace, we subtract from z the part linearly predictable from that subspace's exact matrix coefficients, with the predictor fitted on an independent probe set. We adopted this after a failure. Subspace estimates of near-floor components are noise-dominated, and ablating an estimated subspace spuriously damages the model because the estimate overlaps the dominant directions. For example, the estimated spin-$\frac{1}{2}$ subspace of an $\mathrm{SO}(3)$ model overlaps its spin-1 directions. The function-space ablation is exact for subspaces at the floor, and the quotient conditions supply built-in controls, since the center-orbit average is the identity on quotient models by construction and measures as such (Experiments 4 and 10).

Center discriminations. D_c is the probability that the true product outscores its center translate, D_ω that it outscores both order-three translates, and D_L, D_R, D_{LR} the analogous quantities for the three center elements of $\mathrm{Spin}(4)$. On center-invariant models, the compared scores are equal to numerical precision, and a naive argmax reads spurious wins from floating-point noise. We therefore count winners within 10^{-6} of the maximum and split credit, so an exact tie reads its chance value.

Translation operators. We fit affine maps for left translation ($z(g) \mapsto z(kg)$) and right translation ($z(g) \mapsto z(gk)$) by ridge regression in a PCA frame retaining 99.9% of activation variance, with the regularization strength selected on a validation split, and evaluate R^2, homomorphism defects, and the commutator $\|[M_a^L, M_b^R]\|$ on held-out elements.

Patching. The patching experiments replace the interaction output $h(a, b)$ at inference time by a ridge reconstruction of h from chosen coordinates of a counterfactual product, with the reconstruction fitted on a probe set disjoint from evaluation, and measure how often the model then selects the counterfactual product among 64 candidates.

Character regression and recovery. Raw least-squares regression of the score on characters is unidentifiable under our sampling, because high-Casimir characters are near-collinear near the identity. We therefore regress both the learned score and the exact target on the same sequentially orthogonalized character design in Casimir order and report, per subspace, the model's incremental variance fraction divided by the target's. Standardization and collinearity cancel in the ratio, and a value of 1 means the subspace is carried at its required relative weight. The band-limited calibration in Experiment 15 recovers known support and coefficients exactly, which validates the method.

Ridge probes. All decoding probes select their regularization on a validation split and report on a disjoint test split.

G Experimental record: the mainline $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ program

Each experiment below opens with its purpose, states the protocol, and reports the result. Unless stated, models are the mainline configuration of Appendix C, all numbers are mean ± standard deviation over five seeds, and evaluation is at step 100k.

G.1 Experiment 1: isotypic decomposition and parity diagnostics

The first question is which isotypic subspaces the trained encoder occupies. Table 3 reports the decomposition together with four diagnostics that probe the same distinction independently, namely the cosine between $z(g)$ and $z(-g)$, the sign of the holonomy of z along a 2π rotation, and held-out ridge decoding of the quaternion (an odd, spin-$\frac{1}{2}$ target) and of $qq^\top$ (an even, spin-1 target).

The $\mathrm{SU}(2)$ encoder concentrates in spin $\frac{1}{2}$ (0.835 ± 0.010 at $f = 0.01$, 0.964 ± 0.010 at $f = 0.002$), is odd under the center, reverses sign under a 2π rotation, and decodes the quaternion at $R^2 = 0.995$. The $\mathrm{SO}(3)$ encoder concentrates in spin 1 (0.917 ± 0.008), holds every half-integer subspace at the 0.2% floor, is exactly even with positive holonomy, fails to decode the quaternion ($R^2 \approx 0$), and decodes $qq^\top$ at $R^2 \geq 0.99$. All four diagnostics agree with the decomposition in every seed. Beyond the dominant subspace, the $f = 0.01$ models carry a smaller spin-1 admixture (0.148 ± 0.010) on the $\mathrm{SU}(2)$ side and spin-2 (0.084 at seed 0) on the $\mathrm{SO}(3)$ side, whose role Experiment 3 clarifies.

Robustness across encodings. With the landmark encoding replaced by a frozen random tanh network of the raw coordinates, the pattern reproduces at 60k steps, with $\mathrm{SU}(2)$ spin-$\frac{1}{2}$ at 0.898, quaternion decode $R^2 = 0.993$, and the quotient condition concentrating in spin 1 at 0.876 with the spinor at the floor. With the function-value encoding, $\mathrm{SU}(2)$ carries spin $\frac{1}{2}$ at 0.691 plus spin 1 at 0.214 with quaternion decode $R^2 = 0.990$, and the quotient again pins the spinor at the floor. Across three qualitatively different encodings, the selected irrep is a property of the task and the group.

G.2 Experiment 2: training-fraction sweep and the absence of delayed generalization

Grokking on finite algebraic tasks is a delayed transition from memorization to generalization Power et al. [2022], Nanda et al. [2023]. We test for it across training fractions $f \in \{0.0002, 0.0005, 0.001, 0.002, 0.01\}$, 300k steps, with evaluations every 500 steps.

There is no delayed transition at any f. Every run reaches 90% of its final T2 accuracy within the first 500–1000 steps and is statistically flat for the remaining steps. Lowering f increases the persistent train–test gap, from 0.06 at $f = 0.01$ to 0.27 at $f = 0.0002$ for $\mathrm{SU}(2)$, and only the level of generalization depends on the data budget. Memorization and the group-structured representation coexist rather than compete. At $f = 0.0002$, with 839 training pairs and train accuracy 1.000, the encoder still holds 0.976 of its variance in spin $\frac{1}{2}$ with quaternion decode $R^2 = 0.993$, and 300k steps of weight decay neither remove the memorized component nor improve test accuracy. We attribute the absence of the delay to the smooth structured encoding. The encoder is a continuous function of the observation, so partial generalization is available immediately, and the free-embedding mechanism prior work assigns to the delay is absent. The quotient condition is more data-hungry at every f, with the deficit widening from 4 to 9 points of T2 as f falls.

G.3 Experiment 3: candidate-count sweep and a revised interpretation

More candidates make ranking harder, and the Bayes ceiling falls from 0.951 at $K = 64$ to 0.922 at $K = 256$. We predicted before the sweep that harder ranking would broaden the spin spectrum. The prediction was wrong. At $K = 256$, the $\mathrm{SU}(2)$ encoder concentrates harder (0.975 in spin $\frac{1}{2}$), the score–character fit rises to $R^2 = 0.989$ on the fundamental character alone, fine resolution improves markedly (T3 at $r = 0.2$ rises from 0.618 to 0.921), and held-out accuracy tracks the falling ceiling at a roughly constant 0.96 fraction.

The revision follows from Lemma 1. The fundamental character is monotone in distance to the product, so ranking by it is Bayes-optimal at any resolution, and fine resolution does not require higher-spin content. The spin-1 admixture that accumulates at $K = 64$ coincides with slightly declining fine resolution over training, consistent with drift rather than mechanism. Questions about spectral bandwidth therefore need a regression target, which is the design rationale for Experiments 13–15. The causal suite of Experiment 4 replicates on the $K = 256$ checkpoints, with the spin-$\frac{1}{2}$ ablation collapsing the task to 0.027 and the composition diagnostics strengthening.

G.4 Experiment 4: interventions and the double dissociation

If the model computes with the selected irrep, ablating that subspace should destroy the task and the center discrimination, and ablating anything else of equal rank should do nothing. Both directions hold, and Table 4 reports the full grid. For the $\mathrm{SU}(2)$ model, function-ablating spin $\frac{1}{2}$ collapses ranking to chance (0.026 ± 0.002 against 0.016) and D_c to 0.503 ± 0.006, while ablating spin 1 leaves the task at 1.000 and costs only fine resolution (0.596 to 0.378 at $r = 0.2$, seed 0). For the $\mathrm{SO}(3)$ model the roles transpose exactly, ablating spin 1 is fatal (0.027 ± 0.003) and ablating the absent spin $\frac{1}{2}$ is a no-op. Parity projections agree, and the even projection on the $\mathrm{SO}(3)$ model is an exact identity by construction and measures as one, which validates the intervention machinery. Random rank-matched ablations move nothing. Every cell replicates over five seeds with standard deviations below 0.006.

G.5 Experiment 5: the score is a single character

Regressing the trained score on characters $\chi_j(ab\,c^{-1})$ over triples with candidates at mixed distances, the fundamental character alone explains 0.974 ± 0.006 of score variance for the $\mathrm{SU}(2)$ models at $f = 0.01$ (0.894 ± 0.024 at $f = 0.002$), with no measurable gain from any character up to $j = 3$. For $\mathrm{SO}(3)$, the spinor character explains nothing and χ_1 brings the fit to 0.876 ± 0.014. The learned decision rule is a monotone function of a single faithful character, which is the form Lemma 1 rewards and the compact-group form of the GCR logit.

G.6 Experiment 6: translation operators realize the representation

The claim that activations carry a representation, and correlate with its coefficients, are different claims. We fit affine operators T_k with $z(kg) \approx T_k z(g)$ for Haar-random k and test the algebra they generate. The activations concentrate on 9–14 PCA dimensions holding 99.9% of variance, and the dominant isotypic subspace has exactly the real coefficient-space dimension in every seed, 4 for $\mathrm{SU}(2)$ and 9 for $\mathrm{SO}(3)$, as Lemma 2 prescribes. On that subspace, the fitted operators match the irrep up to a fixed change of basis, measured basis-invariantly. The trace of T_k against $d_j\,\chi_j(k)$ has correlation 1.0000 ± 0.0000 with relative error $1.1 \pm 0.1\%$, and the eigenvalue phases match those of d_j copies of $D^j(k)$ to 0.0014 ± 0.0001 radians for $\mathrm{SU}(2)$ and 0.012–0.021 for $\mathrm{SO}(3)$. Globally, held-out equivariance is $R^2 = 0.90$–0.98, the homomorphism defect $\|T_k T_h - T_{kh}\|$ is 0.04–0.16 relative, and T_e recovers the identity to numerical precision. Left translation acts on the latent space by the representation.

Seeds differ by an intertwiner. Affine stitches between seeds reach held-out $R^2 = 0.96$–0.99 and approximately intertwine the fitted actions, with relative defect 0.07–0.15. Universality across seeds is a statement about a shared representation up to gauge.

G.7 Experiment 7: patching, composition, and the representation ring

If the interaction network implements the composition law, the product state $h(a, b)$ should be controllable through the composed coefficients alone. Replacing $h(a, b)$ by its ridge reconstruction from $D^{1/2}(a'b')$ redirects the $\mathrm{SU}(2)$ model to select $a'b'$ with probability 0.996–0.999 in every seed, equal to or better than patching the full activation (0.92–0.98), while the wrong-irrep reconstruction redirects at chance (0.016 ± 0.008). The eight realified spinor coefficients of the product are causally sufficient to control the output.

Probes agree. The quaternion decodes from h at $R^2 = 0.988$ for the product ab on fresh pairs, against a pair-shuffled control of -1.03, and composing element probes algebraically matches truth at median cosine 0.998. Under the two-variable decomposition of functions on $G \times G$, the state h places $89.9 \pm 1.4\%$ of its variance in the diagonal $(\frac{1}{2}, \frac{1}{2})$ subspace, and $98.2 \pm 0.3\%$ of that energy lies in the 4-of-16-dimensional contraction subspace spanned by the matrix product $D^{1/2}(a)D^{1/2}(b)$. The corresponding $\mathrm{SO}(3)$ numbers are $77.5 \pm 1.6\%$ and $93.9 \pm 0.9\%$ of 9 of 81 dimensions. The hidden layer occupies exactly the channels tensor-generated from the encoder content, pass-through $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ at about 25% each and the bilinear $(\frac{1}{2}, \frac{1}{2})$ at 22%, with all other channels at the floor. The interaction is the specific Clebsch–Gordan contraction that implements the group product, and the network's internal algebra follows the representation ring.

G.8 Experiment 8: the two-sided translation test

Lemma 2 separates full coefficient spaces from every proper subspace. We fit left and right translation operators on six feature systems and evaluate held-out R^2, the right-homomorphism defect, and the commutator defect (Table 7). The trained features pass everything, matching the exact coefficient space at $R^2 = 0.989 \pm 0.001$ per side with commutator 0.012 ± 0.001 ($\mathrm{SU}(2)$, both sides 0.978 ± 0.003, commutator 0.010 ± 0.001). The two-column frame passes the left test exactly and fails right closure at $R^2 = 0.638$, close to the $2/3$ that the Haar-uncorrelated missing column $ge_3 = (ge_1) \times (ge_2)$ predicts. A random truncation at the frame's dimension fails both sides. The raw encoding and a random-initialization encoder fit both sides loosely with commutator defects an order of magnitude above the trained value, so the sharp two-sided closure is learned.

A Schur-type observation makes $\mathrm{SU}(2)$ its own control. The quaternionic spinor's coefficient space has real dimension 4, so its minimal faithful frame coincides with the full space, and the $\mathrm{SU}(2)$ features pass two-sided closure exactly as the lemma predicts. For $\mathrm{SO}(3)$, real type separates the two notions at 6 against 9 dimensions, and the trained model chooses the full space.

G.9 Experiment 9: frame against coefficient space in matched training

Experiment 8 shows the learned features have the closure property. Whether the property is why they win is a training question, so we retrain with the representation content fixed by hand. Two arms share architecture, task, and hyperparameters, and differ only in the encoding, the exact 6-dimensional two-column frame against the exact 9-dimensional coefficient space, both even and standardized, with the encoder trainable so the frame arm is free to reconstruct the missing column nonlinearly. Table 8 reports the result. At statistically equal accuracy (0.891 ± 0.002 against 0.889 ± 0.003), the coefficient arm attains margin 7.29 ± 0.16 against 6.49 ± 0.05 at 17% lower total parameter norm, cheaper in every module, for 31% more margin per unit norm with non-overlapping seed distributions. The frame's saved input coordinates are repaid with interest by the composition cost its one-sided closure forces. This closes the mechanism-selection loop. Two-sided closure makes composition bilinear and scoring linear, bilinearity is cheap under weight decay, and weight decay selects the full coefficient space.

H Experimental record: universality across group forms

H.1 Experiment 10: $\mathrm{SU}(3)$ and $\mathrm{PSU}(3)$

The center of $\mathrm{SU}(3)$ has order three, so the double-cover diagnostic generalizes to a $\mathbb{Z}_3$ discrimination, and real activations track conjugate-pair subspaces since a real feature cannot distinguish an irrep from its conjugate. The $\mathrm{SU}(3)$ encoder places 0.971 ± 0.009 of variance in the fundamental pair $\mathbf{3} \oplus \bar{\mathbf{3}}$, with the center cosine at -0.484 ± 0.005 against the exact triality prediction $\cos(2\pi/3) = -0.5$ and the fundamental coefficients decoding at $R^2 = 0.978 \pm 0.006$. The $\mathrm{PSU}(3)$ encoder places 0.934 ± 0.007 in the adjoint $\mathbf{8}$, the smallest faithful irrep of the quotient, with every charged subspace at the floor and the charged-energy fraction exactly 0.

The interventions repeat the double dissociation in the $\mathbb{Z}_3$ form (Table 5). Orbit averaging collapses the $\mathrm{SU}(3)$ model to chance and is an exact no-op on $\mathrm{PSU}(3)$, removing the invariant part transposes the pattern, and the per-subspace ablations land accordingly. Composition transfers, with fundamental probes composing algebraically at median cosine 0.977. One caveat is on the record. The $\mathrm{PSU}(3)$ model solves its task only partially at this budget (T2 $= 0.685 \pm 0.020$), so the selection statement concerns the representation it builds rather than a solved task, and Appendix K discusses the quotient-hardness pattern.

H.2 Experiment 11: the $\mathrm{Spin}(4)$ family

$\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ has center $\mathbb{Z}_2 \times \mathbb{Z}_2$ and no faithful irrep (Fact 1), and its three central quotients realize different global forms on one Lie algebra. We recorded competing selection hypotheses in the module docstring before running. In two conditions, the minimal-total-dimension choice and the minimal-summand-count choice disagree, which makes them discovery runs.

Table 2: The $\mathrm{SO}(5)$ mixture across conditions (Experiment 12). Vector and adjoint columns are fractions of encoder variance. The mixture survives every condition, and only added data moves it, together with generalization.

condition	T2	vector $\mathbf{5}$	adjoint $\mathbf{10}$
$f = 0.01$, five seeds	0.55–0.64	0.680–0.695	0.211–0.233
200k steps	0.620	0.688	0.214
$K = 256$	0.469	0.664	0.224
weight decay 0.1	0.518	0.660	0.240
$f = 0.03$	0.731	0.707	0.173

The minimal-total-dimension choice wins both. The $\mathrm{SO}(3)\times\mathrm{SU}(2)$ models select $(1,0)\oplus(0,\frac{1}{2})$ with 5 dimensions over the faithful irrep $(1,\frac{1}{2})$ with 6, and the $\mathrm{PSO}(4)$ models select $(1,0)\oplus(0,1)$ with 6 over $(1,1)$ with 9, with the rejected alternatives at 0.0076 ± 0.0002 and 0.0094 ± 0.0011 of variance. The learned representation holds 0.973–0.984 of encoder variance in every condition and every seed. In the $\mathrm{SO}(4)$ condition, the encoding distinguishes (q_L, q_R) from $(-q_L, q_R)$, yet single-factor content is absent ($R^2(q_L) \approx 0$), so only the task group is represented.

The intervention map is summand-resolved (Table 6). Ablating $(\frac{1}{2},0)$ in the $\mathrm{Spin}(4)$ model collapses D_L to 0.507 ± 0.007 with D_R at 1.000 ± 0.000, and the mirror ablation transposes it. The joint discrimination D_{LR} survives either single ablation, since the remaining summand still sees the joint flip. In the $\mathrm{SO}(4)$ condition, only the $(\frac{1}{2},\frac{1}{2})$ ablation matters, and D_{LR} is an exact tie as the quotient requires. In $\mathrm{SO}(3)\times\mathrm{SU}(2)$, the two summands separate causally, with $(0,\frac{1}{2})$ carrying the center charge and $(1,0)$ carrying geometry (task falls to 0.853 ± 0.036 with discriminations intact). Each learned summand is necessary for exactly the center elements its central character detects.

H.3 Experiment 12: $\mathrm{Spin}(5)$, $\mathrm{SO}(5)$, and the precommitted stress test

$\mathrm{Spin}(5) = \mathrm{Sp}(2)$ behaves like every other cover, with the quaternionic spinor $\mathbf{4}$ at 0.964–0.973 of variance and T2 = 0.969–0.989 across five seeds. The quotient does not. The $\mathrm{SO}(5)$ encoder splits between the vector $\mathbf{5}$ and the adjoint $\mathbf{10}$, and before running the replications we precommitted to a decision rule, namely that if the mixture survives replication and conditioning variations it must be reported as a failure of the minimal-representation description rather than explained away.

The mixture survives everything (Table 2). Five seeds at $f = 0.01$ give vector 0.680–0.695 and adjoint 0.211–0.233, and a 200k extension, $K = 256$, and weight decay 0.1 move neither the mixture nor performance in a helpful direction. The one systematic trend is that tripled data raises T2 from 0.55 to 0.731 and lowers the adjoint fraction to 0.173. The mixture is not initialization drift, since the encoding's initial spectrum (vector 0.29, adjoint 0.56) is inverted by step 15k and constant thereafter. Per the rule, the minimal-representation description holds in nine of ten group forms and fails as stated for $\mathrm{SO}(5)$. Two hypotheses remain open. The data trend is consistent with purification as the task approaches a solution, and the cost framing of Experiment 9 makes a genuinely mixed minimum-cost mechanism respectable, since the selected object is the cheapest complete mechanism rather than the smallest representation. Deciding requires a condition that solves $\mathrm{SO}(5)$.

I Experimental record: heat-kernel regression and spectral bias

I.1 Experiment 13: bandwidth is forced and allocated in Casimir order

Ranking never needs more than one character (Experiment 3), so we switch the objective to mean squared error on the standardized heat kernel $K_t(ab\,c^{-1})$ for $t \in \{0.3, 0.1, 0.05\}$, whose envelope $(2j+1)e^{-tj(j+1)}$ requires spectral content that grows as t falls. The single-irrep concentration of ranking is a property of the objective, and the same architecture develops a genuinely multi-spin representation the moment the target requires one. At $t = 0.3$, the encoder spreads over $j \in \{\frac{1}{2}, 1, \frac{3}{2}\}$ and the model reaches test MSE 0.051. At $t = 0.1$ and 0.05, the model fits the lowest-Casimir subspaces the envelope demands, leaves the higher tail unfit, and pays the corresponding

Table 3: Seed statistics for the four principal conditions (Experiments 1, 4, 5, and 10). Five seeds, mean $\pm$ standard deviation, step 100k. D denotes the center discrimination appropriate to the condition, with tie value 0.5 for the $\mathbb{Z}_2$ centers and $1/3$ for $\mathbb{Z}_3$.

	$\mathrm{SU}(2)$	$\mathrm{SO}(3)$	$\mathrm{SU}(3)$	$\mathrm{PSU}(3)$
held-out accuracy (T2)	0.921 ± 0.003	0.883 ± 0.003	0.994 ± 0.001	0.685 ± 0.020
learned-representation FVE	0.835 ± 0.010	0.917 ± 0.008	0.971 ± 0.009	0.934 ± 0.007
forbidden-subspace FVE	0.0023 ± 0.0004	0.0023 ± 0.0004	0.0059 ± 0.0004	0.0000
center cosine	-0.72 ± 0.02	$+1.000$	-0.484 ± 0.005	$+1.000$
coefficient decode R^2	0.995	0.999	0.978 ± 0.006	0.699 ± 0.012
D, intact model	1.0000	0.5000	1.0000	0.3333
score–character R^2	0.974 ± 0.006	0.876 ± 0.014	0.700	

error (0.324 and 0.459 of target variance). The allocation is bottom-up in Casimir order throughout training.

I.2 Experiment 14: what limits the bandwidth

Three manipulations separate the candidate limits. Doubling training time at default width improves test MSE from 0.459 to 0.383 while the spectrum extends to $j = 2$ at 60% of candidate-encoder variance, so the truncation is not a converged optimum. Doubling width at unchanged hyperparameters never escapes the mean predictor (MSE 0.997 through 120k steps), and retuning weight decay to 0.1 lets the wide model train normally and extend its spectrum to $j = 3$ above floor, so the original collapse was a regularization pathology rather than a capacity limit. The retuned wide model lands in a memorization regime (train MSE 0.18–0.24 against test 0.56–0.63). The verdict is a data and regularization trade-off. Width extends the representable spectrum, longer training keeps improving the default model, and extra capacity at this data budget is spent memorizing frozen targets rather than sharpening the generalizing expansion.

I.3 Experiment 15: score-level recovery and the regularization frontier

The activation-level spectra of Experiment 13 leave open whether the score itself is a truncated character expansion. The sequentially orthogonalized recovery analysis of Appendix F answers at the score level, and Table 9 reports the profiles. Recovered subspaces sit near ratio 1, subspaces beyond a sharp truncation sit at exactly 0, and the truncation advances in Casimir order during training, with the $t = 0.05$ spin-2 recovery climbing from 0.12 through 0.83 to 1.20 across 50k–120k steps while $j \geq \frac{5}{2}$ stays at 0.

The band-limited target $\chi_1 + \frac{1}{2}\chi_2$ calibrates the method and isolates the role of regularization. At weight decay 1.0, every out-of-support subspace sits at a 0.0000 floor, and the model prunes the in-support spin-2 remainder, whose 12.6% of target variance matches the final test MSE of 0.131. Partial spin-2 recovery at 25k–50k steps (0.33–0.44) is removed by 60k, so the pruning is an optimization decision rather than a failure to reach the subspace. At weight decay 0.1 on the same target, test MSE falls to 0.004 and the recovery profile is $(1.00, 1.01, 0.94, 0.95)$ with out-of-support floors intact. The truncation frontier is a cost boundary set by regularization, and under weight decay 1.0 the 25-real-dimensional spin-2 coefficient space costs more norm than its variance contribution is worth. The target's central character also controls parity. The band-limited target is center-invariant, and the model on the faithful $\mathrm{SU}(2)$ encoding builds integer spins only, with spin 1 at 0.987 of encoder variance and the spinor at 0.01, so the target's central character overrides the encoding's.

J Summary tables

Figure 2 shows the training dynamics of the isotypic decomposition and the per-group subspace fractions behind Table 1.

K The conjecture, and questions the record raises

Conjecture 1. *In the factorized architecture with weight decay, among parameter settings achieving population ranking loss within ε of the optimum, the minimum-norm setting realizes, as $\varepsilon \to 0$, the*

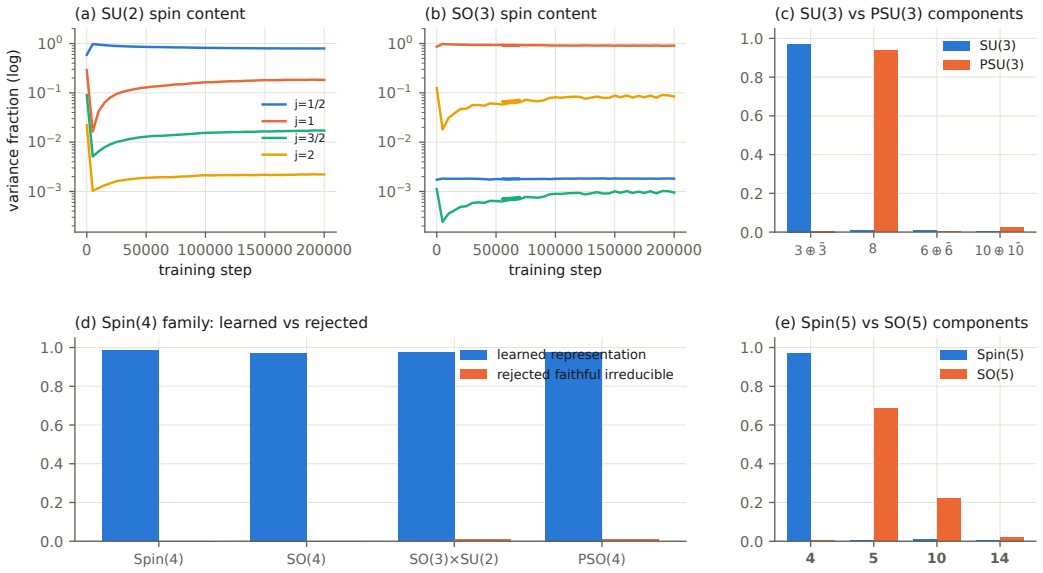


Figure 2: (a, b) Fraction of encoder variance per spin subspace during training for the mainline $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (seed 0, log scale, dotted line at the sampling floor). The two conditions separate within the first evaluations and never cross. (c, d, e) Final subspace fractions for $\mathrm{SU}(3)$/$\mathrm{PSU}(3)$, the $\mathrm{Spin}(4)$ family, and $\mathrm{Spin}(5)$/$\mathrm{SO}(5)$, means over five seeds. Panel (d) includes the rejected faithful alternatives of the two discovery conditions at under 1%.

Table 4: Interventions on the argument encoders for the mainline conditions (Experiment 4). Five seeds, mean $\pm$ standard deviation. Task is ranking accuracy against 63 excluded Haar negatives with chance 0.016, and D_c is the antipode discrimination with tie 0.5. Ablating the selected subspace is fatal, everything else is inert, and the pattern transposes exactly between the conditions.

	$\mathrm{SU}(2)$		$\mathrm{SO}(3)$	
intervention	task	D_c	task	D_c
none	1.000	1.0000 ± 0.0000	1.000	0.5000
ablate spin $\frac{1}{2}$	0.026 ± 0.002	0.503 ± 0.006	1.000 ± 0.000	0.5000
ablate spin 1	1.000 ± 0.000	1.0000	0.027 ± 0.003	0.5000
center-orbit average	0.021	0.498	1.000 (exact identity)	0.5000
random subspace, equal rank	1.000	1.0000	1.000	0.5000

full real coefficient space of a faithful representation minimizing the total regularized cost of features, composition, and scoring, with score a monotone function of $\mathrm{Re}\,\chi_V(ab\,c^{-1})$.

The missing ingredients are a norm lower bound tying ε-optimal scores to the effective dimension of the functions they compute on G, and the exclusion of mechanisms outside the character algebra, the compact-group analogue of the alternative circuits documented for finite groups Stander et al. [2024], Zhong et al. [2023]. Experiments 8 and 9 supply the empirical content, two-sided closure and the cost verdict, and the $\mathrm{SO}(5)$ mixture of Experiment 12 marks the boundary of what the conjecture may claim.

Why do the quotient models generalize worse than their covers? The pattern is uniform, with $\mathrm{SO}(3)$ below $\mathrm{SU}(2)$, $\mathrm{PSU}(3)$ well below $\mathrm{SU}(3)$, and $\mathrm{SO}(5)$ far below $\mathrm{Spin}(5)$, and it widens with the size of the center. We do not yet know the cause. Candidates include information loss and poor conditioning of the averaged encoding near its fold locus, and the intrinsic geometry of the quotient manifold. Until conditioning is matched, with Gram spectra and candidate-hardness diagnostics, the hardness pattern should not be read as a statement about representation selection, and we flag it as the main confound of the quotient program.

Why are some recovery ratios above 1? The $t = 0.05$ profiles in Table 9 overshoot on low spins while the tail is unfit. The model compensates for missing high-Casimir variance by overweighting

Table 5: Interventions for the $\mathrm{SU}(3)$ and $\mathrm{PSU}(3)$ conditions (Experiment 10). Five seeds. D_ω is the probability that the true product outscores both of its center translates, with tie $1/3$.

	$\mathrm{SU}(3)$		$\mathrm{PSU}(3)$	
intervention	task	D_ω	task	D_ω
none	0.998 ± 0.001	1.000	0.739 ± 0.019	0.333
center-orbit average	0.016 ± 0.001	0.337 ± 0.008	0.739 (exact identity)	0.333
remove invariant part	0.998	1.000	0.015 ± 0.001	0.333
ablate $\mathbf{3} \oplus \bar{\mathbf{3}}$	0.018 ± 0.002	0.335	0.733	0.333
ablate adjoint $\mathbf{8}$	0.997	1.000	0.014 ± 0.003	0.333

Table 6: Summand-resolved interventions for the $\mathrm{Spin}(4)$ family (Experiment 11). Five seeds, representative cells with seed statistics. D_L and D_R are the discriminations of the two single-factor center flips, tie 0.5. Each learned summand is necessary for exactly the center elements its central character detects.

condition	intervention	task	D_L	D_R
$\mathrm{Spin}(4)$	none	0.997	1.000	1.000
$\mathrm{Spin}(4)$	ablate $(\frac{1}{2}, 0)$	0.57	0.507 ± 0.007	1.000 ± 0.000
$\mathrm{Spin}(4)$	ablate $(0, \frac{1}{2})$	0.57	1.000	0.498
$\mathrm{SO}(4)$	ablate $(\frac{1}{2}, \frac{1}{2})$	0.017 ± 0.001	0.489	0.489
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	ablate $(0, \frac{1}{2})$	0.53	0.500	0.493 ± 0.006
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	ablate $(1, 0)$	0.853 ± 0.036	0.500	1.000
$\mathrm{PSO}(4)$	ablate $(1, 0)$ or $(0, 1)$	0.809 ± 0.016	0.500	0.500

the subspaces it does carry, and the overshoot relaxes as the next subspace fills (1.81 to 1.55 on $j \leq \frac{3}{2}$ while $j = 2$ climbs). The calibration run shows the analysis reads exact weights when the model carries them, so the overshoot is a property of the truncated solution rather than of the method.

Why is there no grokking? Our encoders are continuous functions of a structured observation rather than free per-element embeddings, and Experiment 2 shows immediate partial generalization at every data budget. This is consistent with accounts that locate the delay in the embedding table. A deliberate attempt to restore the delay, by roughening the encoding kernel or inflating the initialization scale, is a controlled experiment we have not run.

Does the $\mathrm{SO}(5)$ failure infect the other nine forms? The nine successes rest on within-condition replication, exact no-op controls, and interventions that land where central characters predict, and none of that changes. What the failure removes is the claim that minimal-total-dimension selection is a law. The cost picture of Experiment 9 predicts selection of the cheapest complete mechanism, and whether the $\mathrm{SO}(5)$ mixture is that mechanism or a transient of an unsolved task is exactly the open question.

Table 7: The two-sided translation test on the $\mathrm{SO}(3)$ condition (Experiment 8). Five seeds for the trained and random-initialization systems. Left and right columns are held-out R^2 of fitted affine translation operators, hom-R the right-homomorphism defect, and $[L, R]$ the commutator defect, both relative to activation norm. The trained features match the full coefficient space, the frame passes left and fails right, and the random truncation fails both.

feature system	dim	left R^2	right R^2	hom-R	$[L, R]$
trained features	9	0.989 ± 0.001	0.989 ± 0.001	0.012	0.012 ± 0.001
full coefficient space	9	1.000	1.000	0.000	0.000
two-column frame	6	1.000	0.638	0.526	0.039
random subspace of coefficients	6	0.674	0.675	0.411	0.448
raw encoding	32	0.992	0.991	0.058	0.073
random-initialization encoder	16	0.892 ± 0.008	0.890 ± 0.008	0.179	0.160 ± 0.014

Table 8: Frame against coefficient space in matched training on $\mathrm{SO}(3)$ (Experiment 9). Five seeds per arm, 60k steps. Norms are Frobenius norms of module parameters at the end of training. At matched accuracy, the coefficient arm wins margin at lower norm in every module.

encoding	accuracy	margin	encoder norm	interaction norm	scorer norm	margin/norm
frame, 6 dims	0.891 ± 0.002	6.49 ± 0.05	19.6	11.9	13.7	0.243 ± 0.002
coefficients, 9 dims	0.889 ± 0.003	7.29 ± 0.16	16.7	9.8	12.2	0.319 ± 0.008

Table 9: Recovery of the character expansion of the score per checkpoint (Experiment 15). Sequentially orthogonalized profile, model divided by target, with subspaces beyond spin 2 at 0.000 throughout and omitted. The band-limited target is $\chi_1 + \frac{1}{2}\chi_2$. The truncation advances in Casimir order during training, and weight decay 0.1 recovers the band-limited target exactly.

target	step	$j = \frac{1}{2}$	$j = 1$	$j = \frac{3}{2}$	$j = 2$
$K_t, t = 0.3$	25k	1.13	1.12	0.13	0.81
$K_t, t = 0.3$	60k	1.05	1.05	1.01	0.07
$K_t, t = 0.1$	60k	1.47	1.39	1.43	0.00
$K_t, t = 0.05$	50k	1.81	1.81	1.71	0.12
$K_t, t = 0.05$	100k	1.63	1.63	1.49	0.83
$K_t, t = 0.05$	120k	1.58	1.55	1.46	1.20
band-limited, wd 1.0	60k	1.01	1.20	0.00	0.00
band-limited, wd 0.1	60k	1.00	1.01	0.94	0.95